Espressif DSP Library Benchmarks
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The table bellow contains benchmarks of functions provided by ESP-DSP library. The values are CPU cycle counts taken to execute each of the functions. Values in "ESP32" column are for the optimized (assembly) implementation, values in "ANSI C" column are for the non-optimized implementation.

Function name and arguments	CPU cycles	
	ESP32	ANSI C
Dot Product		
dsps_dotprod_f32 for N= 256 points	1057	3879
dsps_dotprode_f32 for N= 256 points, with step 1	1318	4649
dsps_dotprod_s16 for N= 256 points	448	5690
FIR filters		
dsps_fir_f32 1024 input samples and 256 coefficients	1344514	5583242
dsps_fird_f32 1024 samples, 256 coeffs and decimation 4	352282	1425062
FFTs		
dsps_fft2r_fc32 for 64 complex points	5452	11744
dsps_fft2r_fc32 for 128 complex points	12401	26793
dsps_fft2r_fc32 for 256 complex points	28770	60274
dsps_fft2r_fc32 for 512 complex points	62778	135032
dsps_fft2r_fc32 for 1023 complex points	135755	296067
IIR filters		
dsps_biquad_f32 - biquad filter for 1024 input samples	17450	46114
Matrix Multiplication		
dspm_mult_f32 - C[16,16] = A[16,16]*B[16,16];	25695	92827
dspm_mult_s16 - C[16,16] = A[16,16]*B[16,16];	24963	124694
dspm_mult_3x3x1_f32 - C[3,3] = A[3,3]*B[3,1];	81	286

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|dspm_mult_3x3x3_f32 - C[3,3] = A[3,3]*B[3,3];              |       211|       742|
+-----------------------------------------------------------+----------+----------+
|dspm_mult_4x4x1_f32 - C[4,4] = A[4,4]*B[4,1];              |       112|       456|
+-----------------------------------------------------------+----------+----------+
|dspm_mult_4x4x4_f32 - C[4,4] = A[4,4]*B[4,4];              |       404|      1632|
+-----------------------------------------------------------+----------+----------+
```

The benchmark test could be reproduced by executing test cases found in :repo_file:`test/test_dsp.c`.